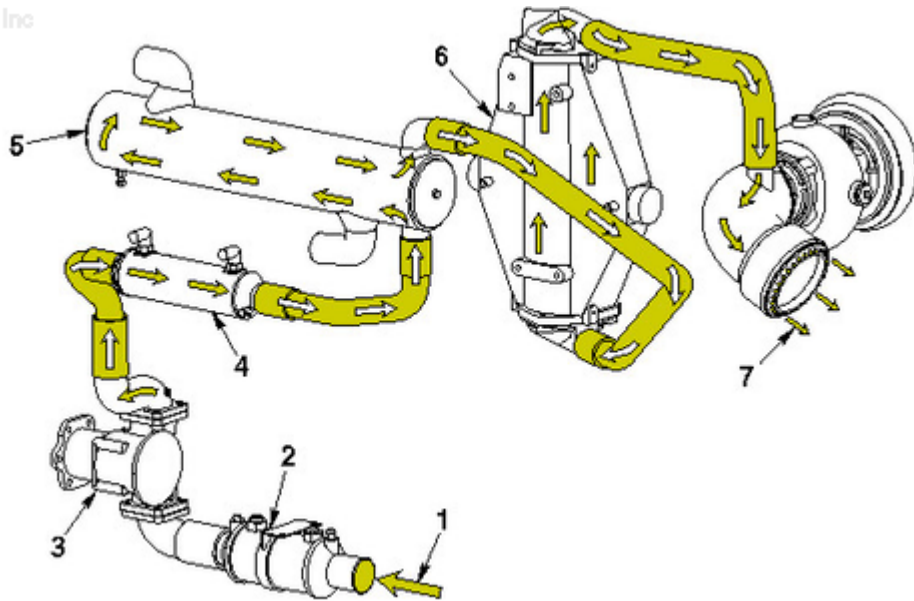


Trainings ▾

Flow Diagram

Marine Applications

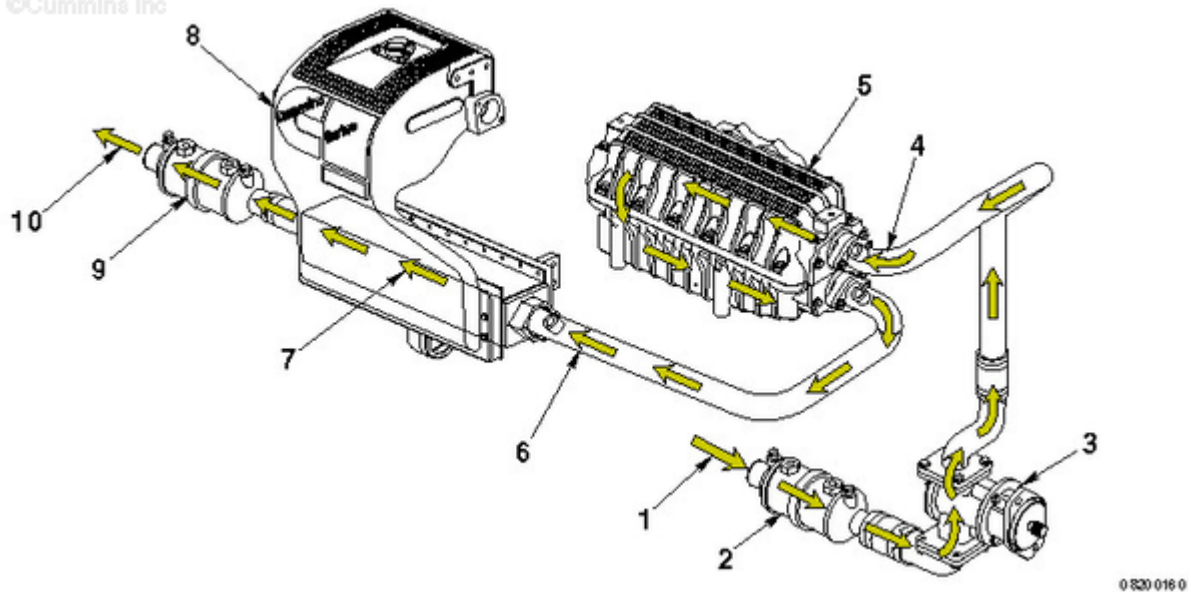
©Cummins Inc



08200242

Sea Water Flow (Heat Exchanger)

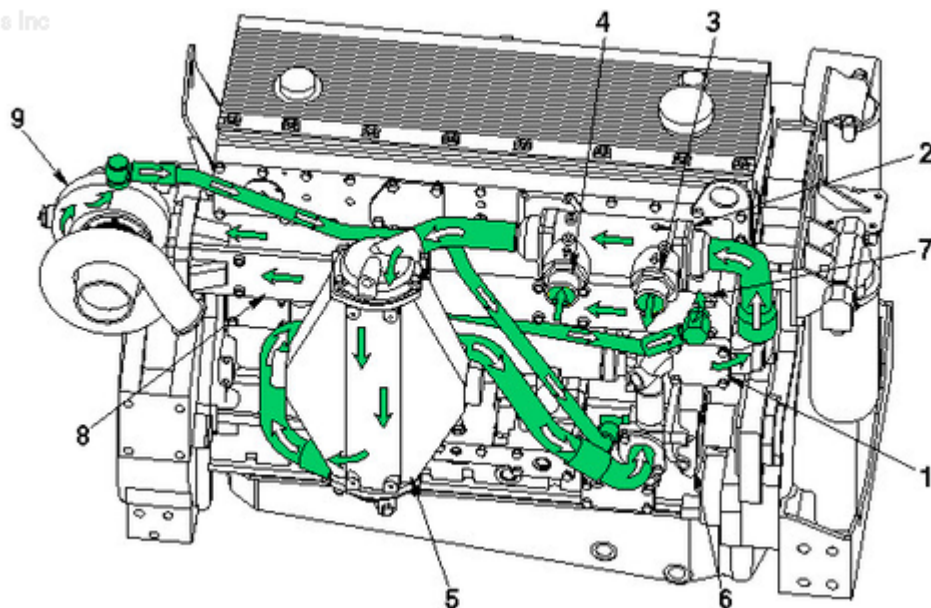
1. Sea water inlet
2. Fuel cooler
3. Sea water pump
4. Gear oil cooler
5. Heat exchanger
6. Aftercooler
7. Sea water outlet.



0 820 016 0

Sea Water Flow (Heat Exchanger)

1. Sea water inlet
2. Fuel cooler
3. Sea water pump
4. Sea water pump to aftercooler
5. Aftercooler
6. Aftercooler to heat exchanger
7. Heat exchanger
8. Expansion tank
9. Gear oil cooler
10. Sea water outlet.



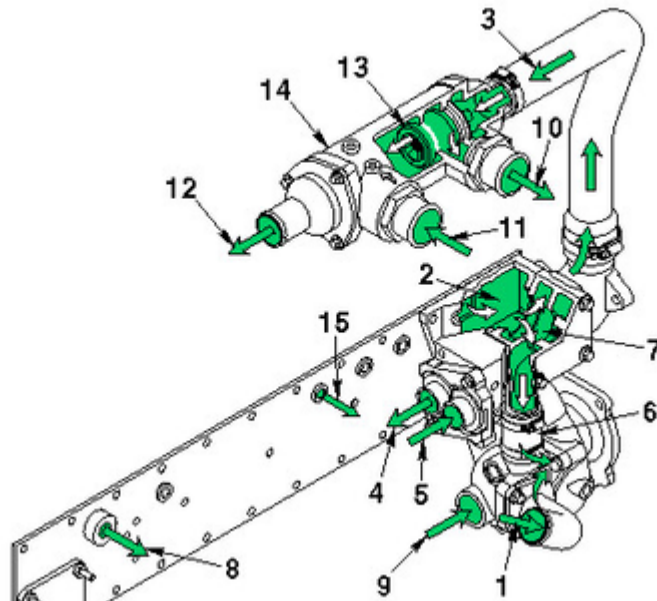
0 820 024 2

Coolant Flow, Keel Cooled (one of three)

1. Engine thermostat cavity (thermostat removed)
2. Keel cooled thermostat housing

3. Coolant out to keel cooler
4. Coolant in from keel cooler
5. Aftercooler
6. Water pump
7. Coolant flow to exhaust manifold
8. Water cooled exhaust manifold
9. Water cooled turbine housing.

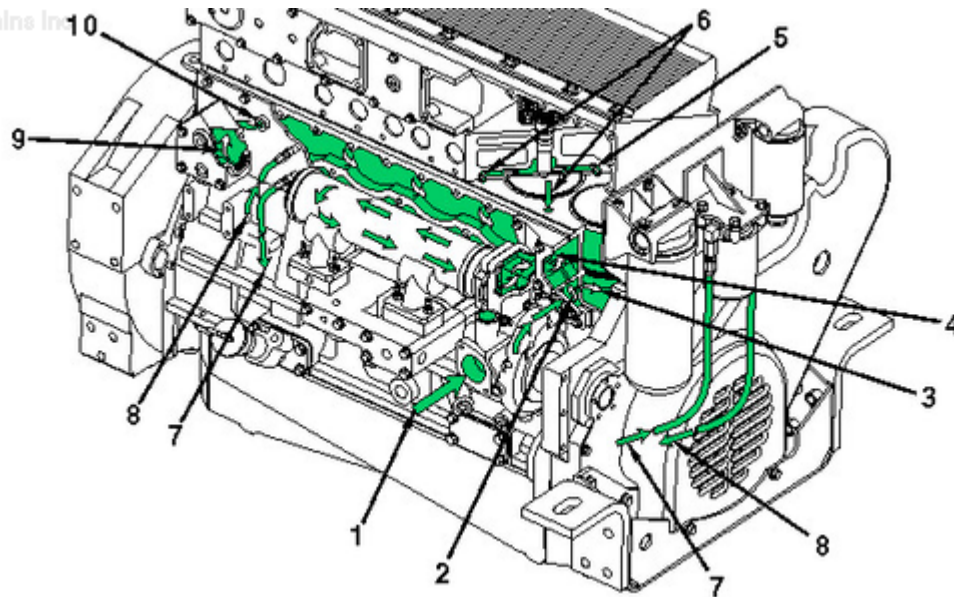
©Cummins Inc



0 820 0240

Coolant Flow, Keel Cooled (two of three)

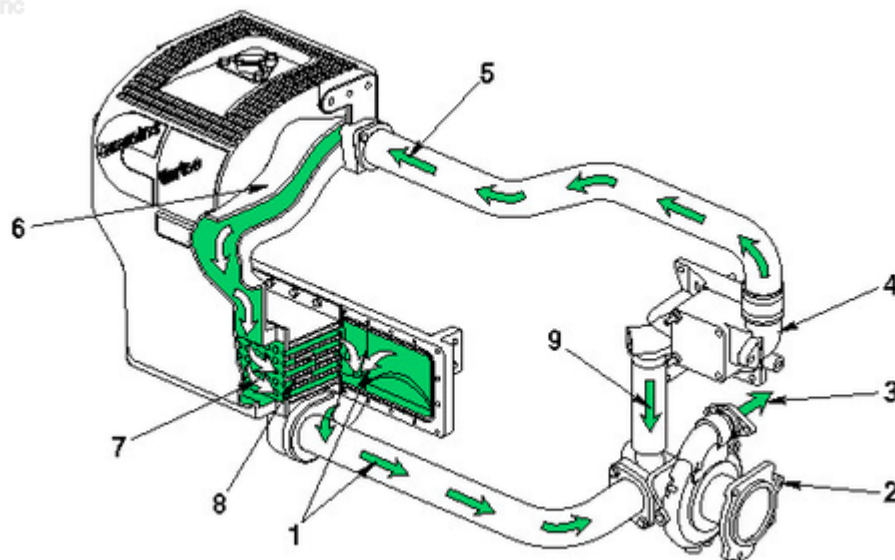
1. Water pump coolant inlet
2. Upper manifold cavity
3. Coolant outlet
4. Lower manifold cavity to cooler
5. Cooler to upper manifold cavity (before thermostat)
6. Coolant bypass hose - plugged
7. Engine thermostat cavity (thermostat removed)
8. To exhaust manifold
9. Turbocharger coolant return to water pump inlet
10. To keel cooler
11. From keel cooler
12. To aftercooler
13. LTA thermostat
14. LTA thermostat housing
15. To turbocharger.



0 820 024 1

Coolant Flow (three of three)

1. Flow to water pump
2. Flow to oil cooler and lower manifold cavity
3. Flow from lower manifold cavity to cylinder liners
4. Flow from oil cooler to thermostat housing
5. Flow to cylinder head
6. Flow from cylinder head to upper manifold cavity
7. Flow to coolant filter
8. Flow from coolant filter
9. Coolant heater - optional
10. Upper manifold to exhaust manifold.

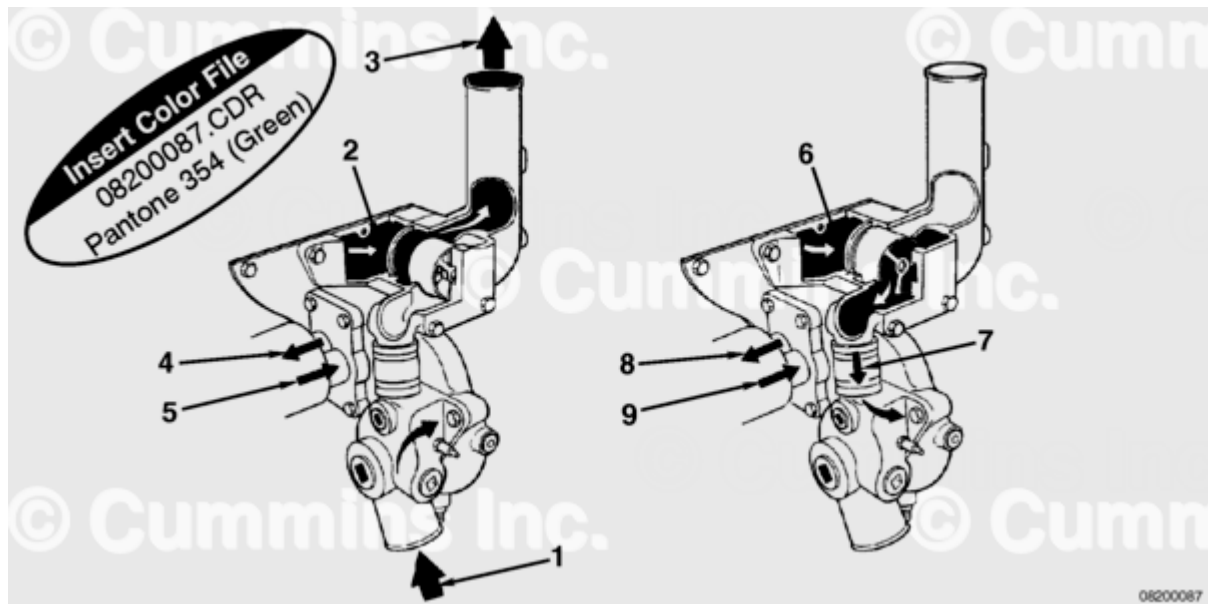


0 820 016 1

Coolant Flow, Heat Exchanger (one of three)

1. Coolant flow to water pump
2. Water pump

3. Coolant flow to lower manifold cavity
4. Thermostat housing
5. Thermostat housing to heat exchanger
6. Expansion tank
7. Heat exchanger sea water core
8. Coolant flow through sea water core
9. Coolant bypass.

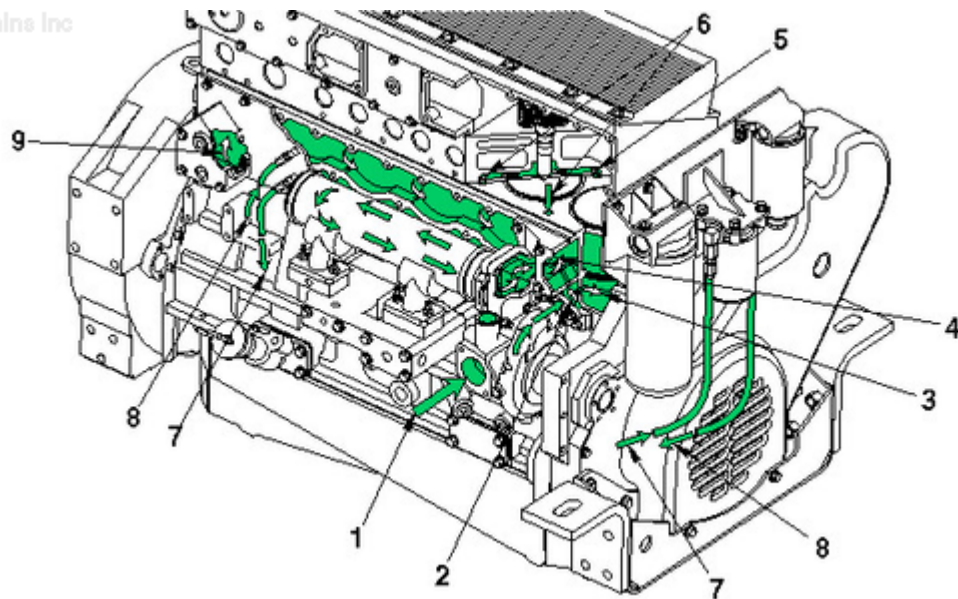


Coolant Flow, Heat Exchanger (two of three)

Thermostat Open (1-5)

Thermostat Closed (6-9)

1. Water pump coolant inlet
2. Upper manifold cavity
3. Coolant outlet
4. Lower manifold cavity to cooler
5. Cooler to upper manifold cavity (before thermostat).
6. Upper manifold cavity (coolant to thermostat)
7. Coolant bypass (return to water pump)
8. Lower manifold cavity to cooler
9. Cooler to upper manifold cavity (before thermostat).



0 820 016 9

Coolant Flow, Heat Exchanger (three of three)

1. Flow from heat exchanger to water pump
2. Flow to oil cooler and lower manifold cavity
3. Flow from lower manifold cavity to cylinder liners
4. Flow from oil cooler to thermostat housing
5. Flow to cylinder head
6. Flow from cylinder head to upper manifold cavity
7. Flow to coolant filter
8. Flow from coolant filter
9. Coolant heater - optional.